Apache Web Server Administration

Session 6



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Objectives

- *Describe the administration of Apache Web Server using logs*
- *Describe the configuration logs in Apache*
- *Explain the contents in the log files*
- *Explain the creation of the custom error pages*

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Types of Logs

- Error Log
  - Records errors
  - Records the diagnostic messages
- Access Log
  - Records all the requests received by the server
- PID File
  - Records the process ID of the parent httpd process
- Script Log
  - Records the input to and the output from the CGI scripts
- Piped Log
  - Enable log rotation without restarting the Web server

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- The `LogLevel` directive defines the type of the error

Syntax

```
LogLevel level
```

where,

`level` - specifies the severity of the error

Snippet

```
LogLevel crit
```

Here Apache records:

- Critical condition errors
- Errors with `emerg` and `alert` importance

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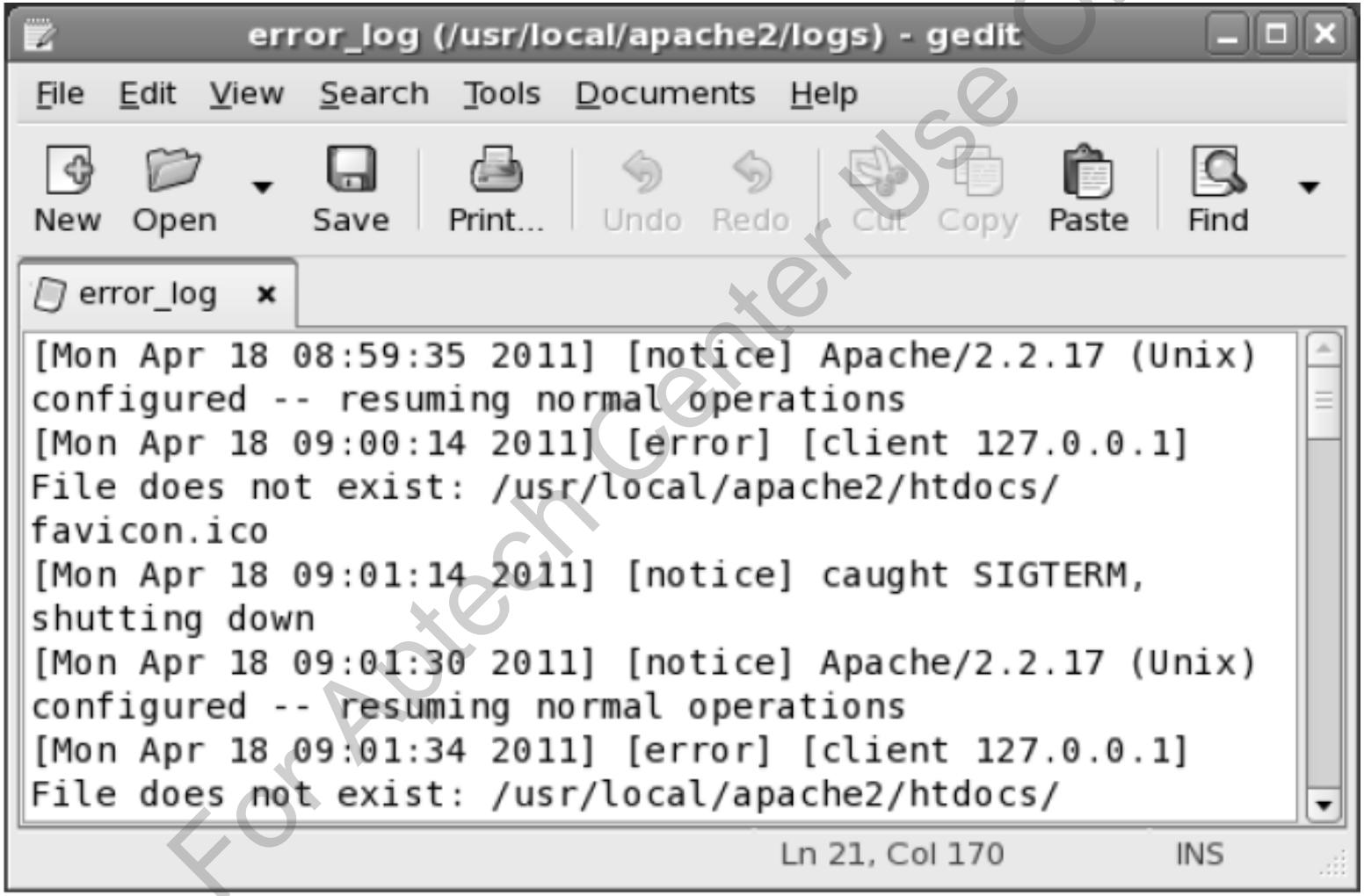
- Error Level:

Level	Severity of the Error
emerg	Indicates that the system is unusable
alert	Indicates that immediate action must be taken
crit	Indicates a critical condition
error	Indicates a non-critical error condition
warn	Indicates a warning condition
notice	Indicates normal but significant condition
info	Indicates an informational message
debug	Indicates a debug level message

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- To view the `error_log` file, perform the following steps:
  1. Using the GUI in Linux, browse to the `/usr/local/apache2/logs` directory
  2. Open the `error_log` file using the `gedit` text editor

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Configuring Access Logs

- Transfer Log Directive specifies the name and location of a log file

Syntax

```
TransferLog filepath
```

Snippet

```
TransferLog /usr/local/apache2/log/access_log
```

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Configuring Log File content

- The `CustomLog` and `LogFormat` directives are used to configure the contents of the `access_log` file
- The two types of log format are as follows:
  - Common Log Format
  - Combined Log Format

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Common Log Format

- The `LogFormat` directive can be used to modify the standard log format

Syntax

```
LogFormat format-string nickname
```

Where

`format-string` - specifies the percent directives that indicates a distinct piece of information to be logged in the access_log file

`nickname` - specifies the name assigned to the format string. Defining the nickname is optional

Snippet

```
LogFormat "%h %u %t \"%r\" %>s %b"
```

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- Commonly Used percent directives:

Percent Directive	Description
%{Header }i	Specifies an incoming HTTP header in the client request
%{Header }o	Specifies an outgoing HTTP header in the server response
%{Variable}e	Specifies an environment variable as defined by the server
%b	Specifies the file size delivered by the server, excluding HTTP headers
%c	Specifies the connection status
%D	Specifies the time the server took to process the request
%f	Specifies the file path of the requested document
%h	Specifies the hostname of the client. If the hostname cannot be resolved, the IP address of the client is logged
%H	Specifies the protocol used for the request
%m	Specifies the request method
%p	Specifies the TCP port number on which the request arrives
%s	Specifies the HTTP status code
%t	Specifies the time of the request
%T	Specifies the number of seconds taken by the server to process the request
%u	Specifies the remote user in authenticated requests
%U	Defines the requested URL

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- The `CustomLog` directive defines the format and the filename of the log file

Syntax

```
CustomLog file|pipe format|nickname
```

Where,

`file` - specifies the location and name of the log file relative to the server root

`pipe` - specifies the path for the program that receives the input log information

`format` - specifies the log format. It can be an explicit format string

`nickname` - specifies a nickname already defined by a previous LogFormat directive

Snippet

```
LogFormat "%h %t \"%r\" %>s" common
CustomLog logs/access_log common
```

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Combined Log Format

Syntax

```
LogFormat format-string nickname
```

Snippet

```
LogFormat "%h %l %u %t \"%r\" %>s %b \"%{Referrer}i\"
\"%{User-agent}i\"" combined
CustomLog log/access_log combined
```

`Referrer` specifies the site name and `Useragent` specifies the browser used by the client to access the site.

The `referrer` information is stored in the `referrer_log` file and the `user-agent` information is logged to the `agent_log` file.

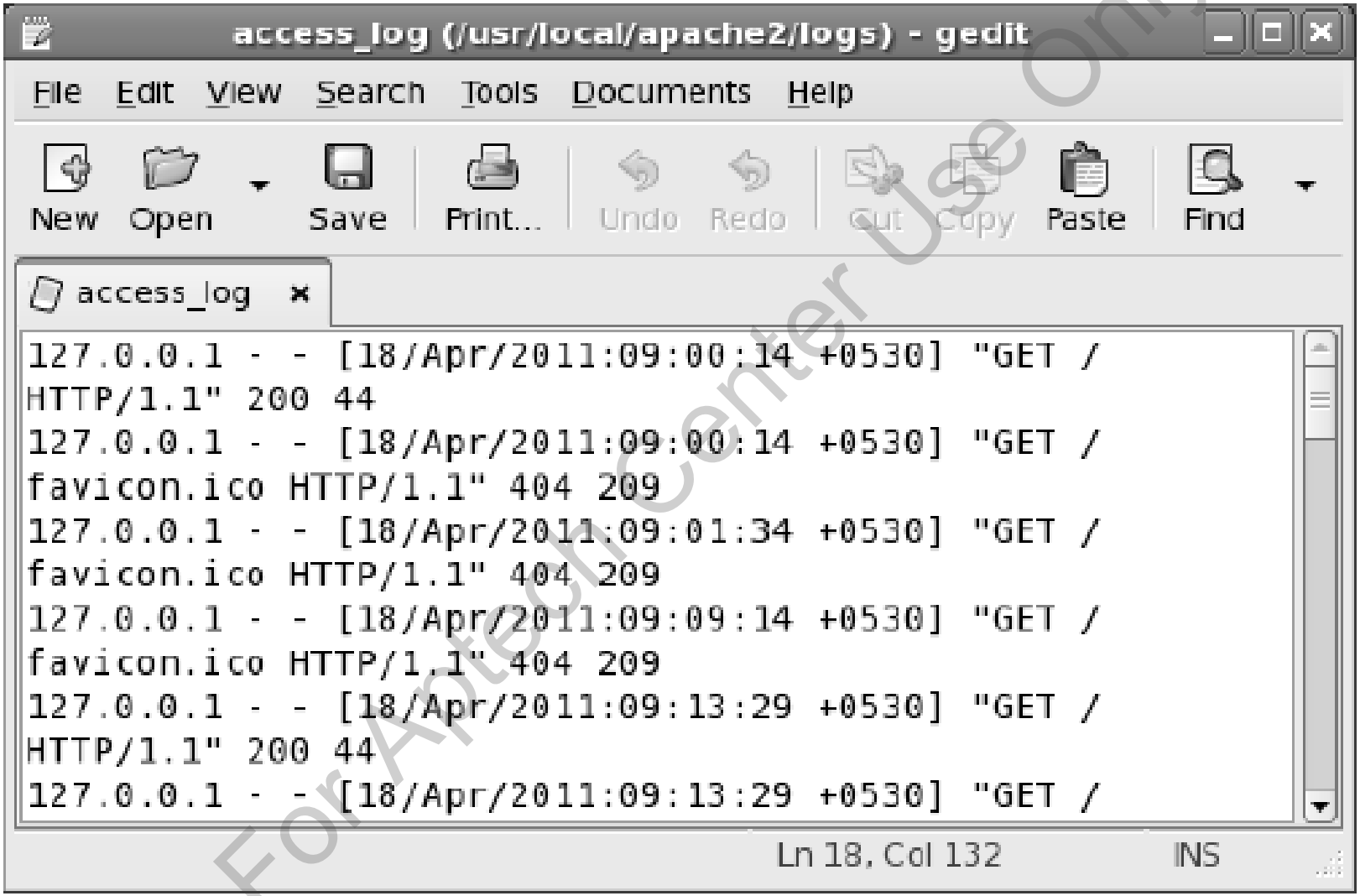
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View Access Log

1. Using the GUI in Linux, browse to the `/usr/local/apache2/logs` directory
2. Open the `access_log` file using the `gedit` text editor

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View Access Log



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- To create a custom error page:
  1. Create a new directory named error under the `/usr/local/apache2/htdocs` directory
  2. Open the `gedit` text editor
  3. Enter the code as shown

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Custom Error Page

```
<html>
  <head>
    <title> We’re sorry, the page you requested could not be found
</title>
  </head>
  <body>
    <font face = “arial” size=”-1”>
    <h1 align = “left”> Custom Error Page </h1>
    <hr align = “left” width=”100%”>
    <p><b>  We’re  sorry,  but  there  is  no  page  matching  your
request.</b>   </p>
    <p>
The page you are looking for might have been removed, had its name
changed, or is temporarily unavailable. It is also possible that you
typed the address incorrectly. </p>
    <p>You can do one of the following...</p>
    <ul>
```

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Custom Error Page

```
    <li>
      Open the <a href="/"> home page </a>, and then look for links to the
information you want.
    </li>
    <li>
      Click the <A href = "javascript:history.back(1)"> Back </A> link to try
another link.
    </li>
    <li>
      If you typed the page address in the Address bar, make sure that it is
spelled correctly
    </li>
  </ul>
  </p>
    <hr align="left" width="100%">
    </font>
  </body>
</html>
```

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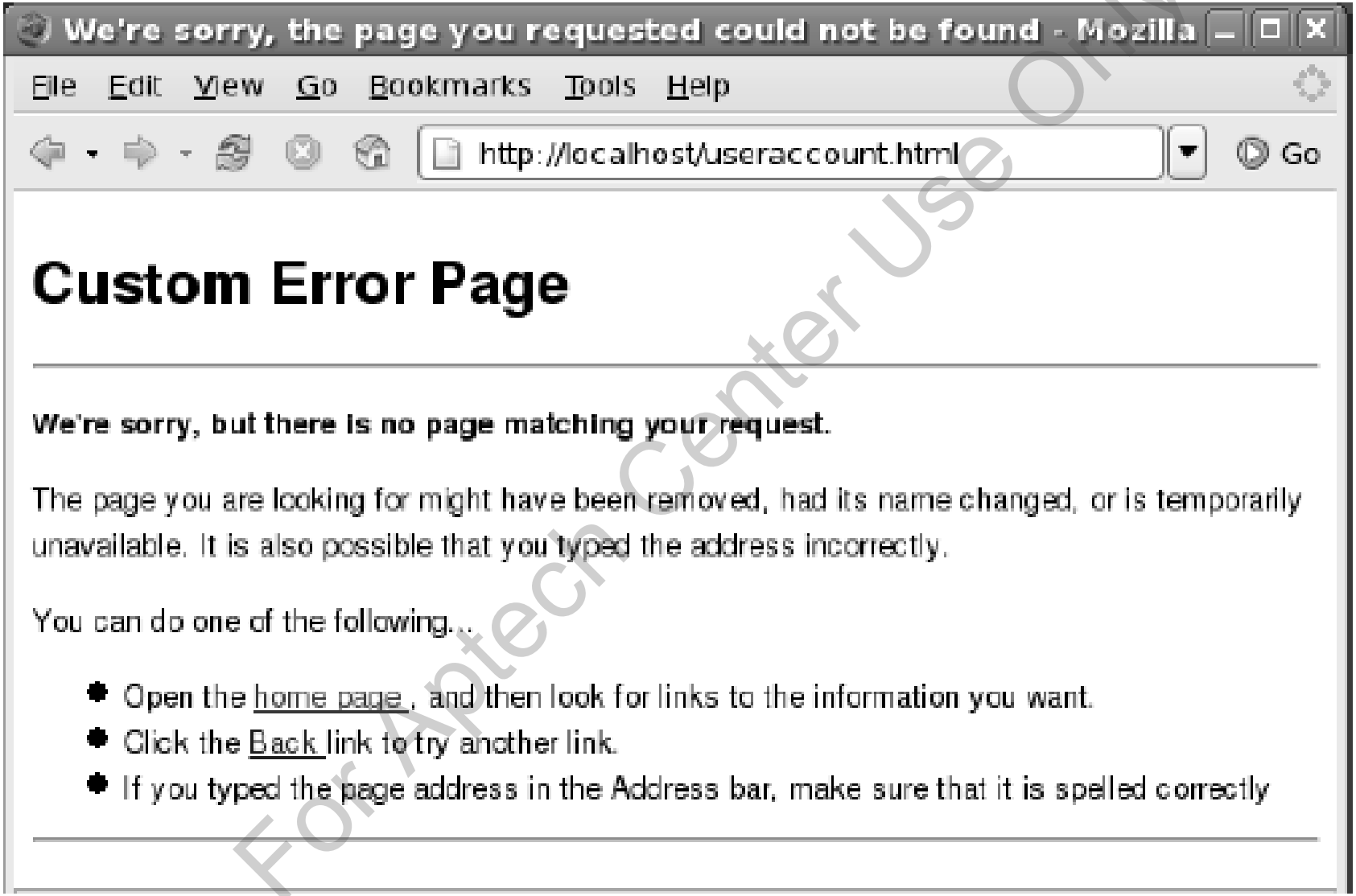
4. Save the file as 404.html in the */usr/local/apache2/htdocs/error/* directory
5. Open the *httpd.conf* file using the `gedit` text editor
6. Enter the following code:

```
ErrorDocument 404 /error/404.html
```

7. Save the `httpd.conf` file and restart Apache Web server
8. Open the Mozilla Firefox Web browser
9. Access a nonexistent html page, such as:

```
http://localhost/useraccount.html
```

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Summary

- The two main log files in Apache server are `error_log` and `access_log`
- The `error_log` file contains information about the errors that are generated during the operation of the server
- The `access_log` file records information about the requests received by the server
- Apache can be configured to display custom error pages when an error is encountered
- The `ErrorDocument` directive specifies the custom error page associated with a particular error code

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